

Chapters 25 and 26: Diagonalization and similarity Worksheet

1. Are each of the following statements true or false?

- (a) If A is diagonalizable and 0 and 1 are the only eigenvalues of A , then $A^2 = A$.
- (b) If A is a 3×3 matrix with only one distinct eigenvalue, A is not diagonalizable.
- (c) If A is a 4×4 matrix with characteristic equation given by $(\lambda^2 - 4)(\lambda + 1)(\lambda - 3) = 0$, then A must be diagonalizable.
- (d) All diagonalizable matrices are also invertible.
- (e) All invertible matrices are also diagonalizable.
- (f) If A is a 4×4 matrix with characteristic equation given by $(\lambda - 5)^3(\lambda + 2) = 0$, then A will not be diagonalizable.
- (g) If A is invertible, then A is similar to the identity matrix.

2. Determine if each of the following matrices is diagonalizable. NOTE: you worked out the eigenvalues/eigenvectors for these matrices on the previous worksheet, so most of the hard work is done. You can request for your TA to write the eigenvalues/eigenvectors on the board.

(a) $A = \begin{bmatrix} -4 & -6 \\ 3 & 5 \end{bmatrix}$

(b) $A = \begin{bmatrix} 4 & 2 \\ -2 & 8 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & -5 & 0 \\ 1 & 7 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(d) $A = \begin{bmatrix} -2 & 5 & 5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$

3. Consider $A = \begin{bmatrix} -4 & 6 \\ -3 & 5 \end{bmatrix}$. Diagonalize A (that is, find P , D and P^{-1}), and use your diagonalization to compute A^8 .

4. Use the definition of *similar* to prove each statement. Use full sentences.

- (a) If A is similar to B , then they have the same determinant.
- (b) The only matrix similar to I_n is I_n . In other words, if C is a matrix that is similar to I_n , then C must be equal to I_n .

5. Is $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 1 & -3 & 4 \end{bmatrix}$ diagonalizable?

For extra practice, diagonalize A (Find P, P^{-1}, D).

6.

(a) Consider $A = \begin{bmatrix} 4 & -16 \\ 1 & -4 \end{bmatrix}$. Is A invertible? Is A diagonalizable?

(b) Consider $B = \begin{bmatrix} 1 & -2 \\ -4 & 8 \end{bmatrix}$. Is B invertible? Is B diagonalizable?